

Course Title: Artificial Intelligence

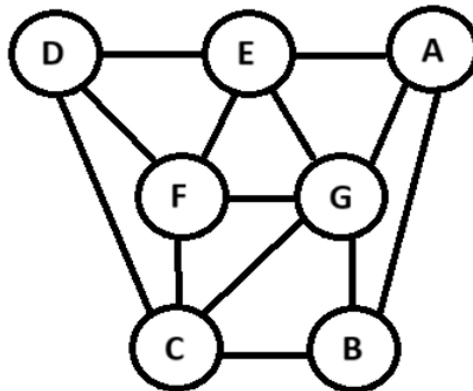
Term: Spring 2025

Quiz Title: Quiz 1 – Search Algorithms

Time: 45 minutes

Question 1 (tree search, graph search): (12 pts)

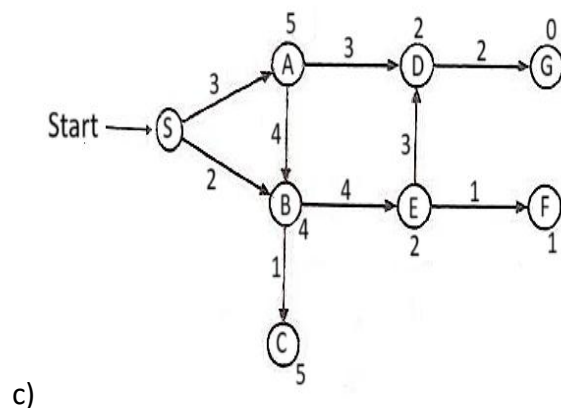
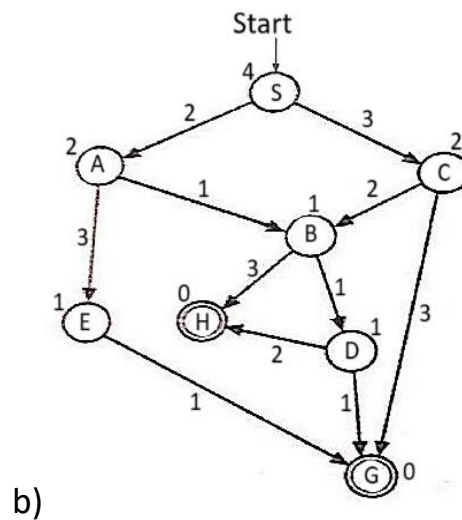
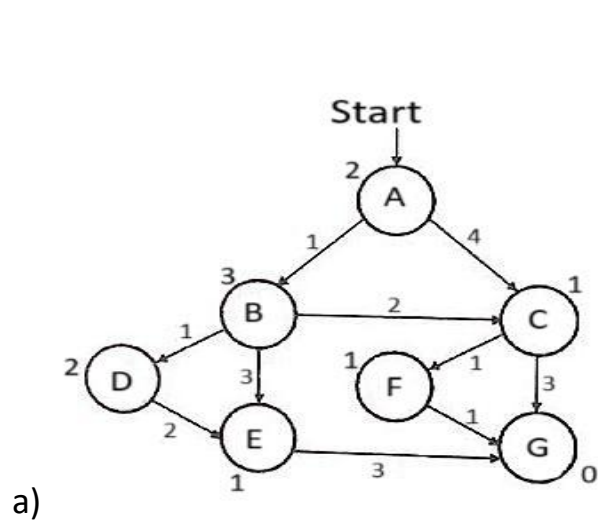
For the following graph, using depth-first search with both tree search and graph search approaches, state the traversal order of the nodes. Start from node D.



Question 2 (A*): (18 pts)

For each graph, draw the search tree of the A* algorithm for both goal test at generation time and goal test at expansion time, and specify the path found. The numbers on the edges are the path cost between two nodes, and the numbers next to the nodes are the estimated cost to the goal. The order of visiting the children of each node is in alphabetical order. The estimated cost at the goal node is zero.

If you consider any assumptions for the solution, make sure to mention them in your answer.



Question 3 (BFS): (30 pts)

Suppose that for a problem with breadth-first search and goal testing at generation time, 32 nodes need to be expanded. If the branching factor of the tree is fixed and the depth of the tree is 5, and the goal is at depth 4, find the allowed range of the branching factor. The root is at depth 0.

Question 4 (BFS): (40 pts)

In an infinite two-dimensional grid, each cell is connected to its 4 neighbors (up, down, left, and right). The starting point of the search is point (0,0) and the goal is at position (X, Y). In two search modes, without checking for repeated states and with checking for repeated states, how many nodes does BFS algorithm expand before reaching the goal?

Good luck!